Don't fear the unlabelled: safe semi-supervised learning via simple debiasing

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TL;DR: We propose a simple modification of SSL methods to make them safe by debiasing their risk estimate.

Semi-supervised learning.

- Labelled data are expensive and limited: $\{(x_i, y_i)\}_{i\in\mathcal{L}}$
- Unlabelled data are cheap and plentiful: $\{x_i\}_{i\in\mathcal{U}}$

Missing Completely At Random (MCAR) i.e. y being missing is independent of the data (x, y).

Goal: $\theta^* \in \arg\min_{\theta\in\Theta} \mathcal{R}(\theta) = \arg\min_{\theta\in\Theta} \mathbb{E}[L(\theta; x, y)]$

Motivations: Safe semi-supervised learning

- Critical application of SSL on real-world datasets (not classical SSL benchmark: CIFAR, SVHN, STL10...) [Oliver et al., NeurIPS 2018]
- There is **no best** SSL methods [USB, Wang et al., NeurIPS 2022]:
- **No theoretical guarantees** without strong assumptions on the data distribution [Mey and Loog, PAMI 2022]

Contribution: Competitive methods can be easily modified to benefit from theoretical guarantees without performance degradation.

Simplest baseline: The complete case

$$\hat{\mathcal{R}}_{CC}(\theta) = \frac{1}{|\mathcal{L}|}\sum_{i\in\mathcal{L}} L(\theta; x_i, y_i)$$

$\longrightarrow$ Under MCAR unbiased $\Rightarrow$ learning theory + asymptotic statistics.

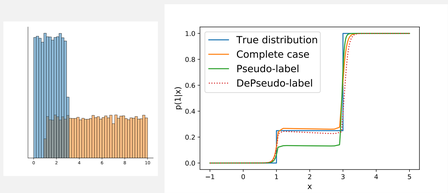
Safeness: A SSL algorithm is safe if it has theoretical guarantees that are similar to or stronger than the complete case baseline.

Including unlabelled data

$$\hat{\mathcal{R}}_{SSL}(\theta) = \frac{1}{|\mathcal{L}|}\sum_{i\in\mathcal{L}} L(\theta; x_i, y_i) + \frac{\lambda}{|\mathcal{U}|}\sum_{i\in\mathcal{U}} H(\theta; x_i)$$

- Entropy minimisation (Grandvalet and Bengio,2004):
$$H(\theta; x) = -\sum_{k\in\mathcal{C}} f_\theta(x)_k \log(f_\theta(x)_k).$$
- Consistency-based (UDA, VAT, Π-Model ...):
$$H(\theta; x) = Div(f_{\hat{\theta}}(x), f_\theta(pert(x))).$$
- Pseudo-label based (Fixmatch and Fixmatch related):
$$H(\theta; x) = -\log(\max f_\theta(pert(x))_k)\mathbb{1}(f_\theta(x)_k > \tau).$$

SSL failure on a toy example



- **Biased** risk estimator $\Rightarrow$ learning theory does not hold.
- No **consistency** $\Rightarrow$ fail even with an infinite amount of labelled data points

DeSSL: Unbiased under MCAR

$$\hat{\mathcal{R}}_{DeSSL}(\theta) = \frac{1}{|\mathcal{L}|}\sum_{i\in\mathcal{L}} L(\theta; x_i, y_i) + \frac{\lambda}{|\mathcal{U}|}\sum_{i\in\mathcal{U}} H(\theta; x_i) - \frac{\lambda}{|\mathcal{L}|}\sum_{i\in\mathcal{L}} H(\theta; x_i).$$

Is $\hat{\mathcal{R}}_{DeSSL}(\theta)$ an accurate risk estimate?

Theorem 1 *The function $\lambda \mapsto Var(\hat{\mathcal{R}}_{DeSSL}(\theta)|r)$ reaches its minimum for:*

$$\lambda_{opt} = \frac{|\mathcal{U}|}{|\mathcal{L}| + |\mathcal{U}|} \frac{\mathrm{Cov}(L(\theta; x, y), H(\theta; x))}{Var(H(\theta; x))}$$

and

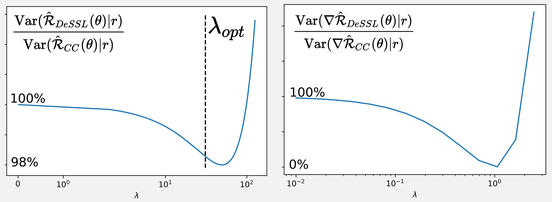
$$Var(\hat{\mathcal{R}}_{DeSSL}(\theta))|_{\lambda_{opt}} = \left(1 - \frac{|\mathcal{U}|}{|\mathcal{L}| + |\mathcal{U}|}\rho^2_{L,H}\right) Var(\hat{\mathcal{R}}_{CC}(\theta)) \leq Var(\hat{\mathcal{R}}_{CC}(\theta)).$$

where $\rho_{L,H} = Corr(L, H)$.

Remark 1: Heuristic for H: H and L should be as correlated as possible.

Remark 2: Debiasing with the labelled set $\mathcal{L}$ is optimal.

Variance reduction



Estimate of the variance reduction on CIFAR-10's test set using a pre-trained CNN-13.

DeSSL is safe:

Calibration *If the original loss is a proper scoring rule, then DeSSL is also a proper scoring rule.*

Consistency *Under usual regularity conditions of M-estimators for both L and H, $\hat{\theta} = \arg\min \hat{\mathcal{R}}_{DeSSL}$ is asymptotically consistent.*

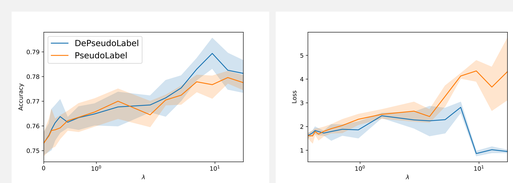
Generalisation error bounds *If L and H are bounded, DeSSL benefits from generalisation error bounds based on the Rademacher complexity.*

Asymptotic normality *Under usual conditions on L, H and the risk $\mathcal{R}(\theta)$, DeSSL is asymptotically normal. Its asymptotic variance can be optimised in λ and is smaller than the complete case's variance at its optimum.*

Interpretation of DeSSL

- H **is a Control variate** (variance reduction technique)
- $\hat{\mathcal{R}}_{DeSSL}(\theta)$ is the Lagrangian of the following optimisation problem:
$$\min_\theta \quad \hat{\mathcal{R}}_{CC}(\theta)$$
$$\text{s.t.} \quad \frac{1}{|\mathcal{L}|}\sum_{i\in\mathcal{L}} H(\theta; x_i) = \frac{1}{|\mathcal{U}|}\sum_{i\in\mathcal{U}} H(\theta; x_i)$$
- H acts like a regularisation of the model's confidence on the labelled

Pseudo-label on CIFAR-10 with various λ ($|\mathcal{L}| = 4000$)



Accuracy (Left) Cross-entropy (Right).

DeFixmatch on CIFAR-10 ($|\mathcal{L}| = 4000$)

	Complete Case	Fixmatch	DeFixmatch
Accuracy	87.27 ± 0.25	93.87 ± 0.13	**95.44 ± 0.10**
Worst class accuracy	70.08 ± 0.93	82.25 ± 2.27	**87.16 ± 0.46**
Cross-entropy	0.60 ± 0.01	0.27 ± 0.01	**0.20 ± 0.01**
Brier score	0.214 ± 0.005	0.101 ±0.003	**0.076 ± 0.001**

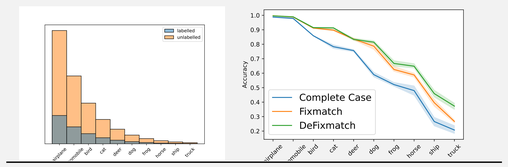
Disparate effect of SSL ($|\mathcal{L}| = 4000$)

	Complete Case	Fixmatch		DeFixmatch	
	Accuracy	Accuracy	$\mathcal{BR}$	Accuracy	$\mathcal{BR}$
airplane	86.94	95.94	0.88	96.62	0.94
automobile	95.26	97.54	0.68	98.22	0.89
bird	**80.46**	**90.80**	**0.68**	**92.64**	**0.80**
cat	**70.08**	**82.50**	**0.56**	**87.16**	**0.78**
deer	88.88	95.86	0.78	97.26	0.94
dog	**79.66**	**87.16**	**0.53**	**90.98**	**0.81**
frog	93.12	97.84	0.80	98.62	0.94
horse	90.96	96.94	0.83	97.64	0.92
ship	94.12	97.26	0.67	98.06	0.84
truck	93.18	96.82	0.84	97.20	0.93

Benefit ratio, $\mathcal{BR}$ = the impact of SSL on a class (Zhu et al., ICLR 2022).

Long-tailed SSL: CIFAR-10-LT ($|\mathcal{L}| = 20\%$)

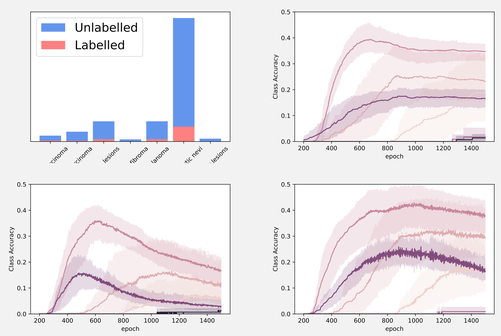
Unbalanced MCAR dataset based on CIFAR10 [ABC, Lee et al., NeurIPS 2021].



	Complete Case	Fixmatch	DeFixmatch
Accuracy	64.29 ± 0.82	72.95 ± 0.63	**76.10 ± 0.74**
Worst class accuracy	20.72 ± 3.07	26.66 ± 2.36	**37.26 ± 2.44**

DePseudo-label on TissueMNIST ($|\mathcal{L}| = 10\%$)

Yang et al., Nature Scientific Data, 2023



Classes distribution (Top Left). Accuracy per class through training Complete Case (Top Right) PseudoLabel (Bottom Left) DePseudoLabel (Bottom Right).